

Hackathon Problem Statements

Regd No: 25091A3264, 25091A32A2

Problem Statement No: 5

Pesticide Usage Optimizer Build a decision-support tool that minimizes pesticide cost while maximizing crop protection using dynamic programming to solve a constrained optimization problem across multiple pest threats and budget limits. Integrate historical pest-outbreak data with weather patterns to predict pest risk windows using classification models. Design pesticide and crop objects using OOP inheritance to handle different chemical compatibility rules. This reduces chemical overuse, protecting both soil health and farmer expenditure.

Regd No: 25091A3256, 25091A3253

Problem Statement No: 26

Hospital Readmission Predictor Build a predictive model using decision trees or random forests trained on de-identified electronic health record patterns to estimate a patient's risk of readmission within 30 days of discharge. Identify key contributing factors (comorbidities, follow-up compliance) to help hospitals design better discharge planning. Use OOP to model patient history and treatment episodes as linked objects. This can reduce healthcare costs and improve patient outcomes in resource-constrained public hospitals.

Regd No: 25091A32K1, 25091A32K7

Problem Statement No: 47

Noise Pollution Mapping System Create a noise pollution mapping tool that aggregates simulated or public sensor data across a city into a spatial heatmap using statistical aggregation methods, identifying noise violation hotspots near hospitals and schools. Apply time-series analysis to identify peak noise periods by location. Use OOP to model sensors, zones, and permissible-limit rules. This supports urban planning bodies in enforcing noise pollution control regulations more effectively.

Regd No: 25091A3244, 25091A3210

Problem Statement No: 55

Disaster Shelter Capacity & Allocation Optimizer Create a shelter allocation system using bipartite matching algorithms to assign displaced families to the nearest available shelters with sufficient capacity during floods or other disasters. Use OOP to model shelters, families, and resource needs (food, medical) as connected entities. This ensures fair, efficient, and rapid shelter allocation during crisis situations when manual coordination breaks down.

Regd No: 25091A3209, 25091A32E0

Problem Statement No: 119

Post-Disaster Infrastructure Damage Assessment Classifier Design an infrastructure damage classification tool using classification algorithms on structured damage-report data (or basic image features) to categorize damage severity across roads, bridges, and buildings after a disaster. Use OOP to model infrastructure assets and damage records. This speeds up damage assessment surveys that traditionally take weeks of manual inspection, accelerating relief fund allocation.

Regd No: 25091A32D1, 25091A32D0

Problem Statement No: 127

RTI (Right to Information) Request Routing & Tracking System Build an RTI management system using priority queues to track pending requests against statutory response deadlines, automatically flagging overdue requests for escalation. Use OOP to model requests, departments, and compliance status. This improves accountability in RTI processing, which frequently suffers from delays despite legal response-time mandates.